A Paired, Floor-Guarded Evaluation Protocol for TensorRT INT8 and FP8 Object Detectors under Synthetic Image Corruptions

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ABSTRACT

Robustness comparisons of quantized detectors often use the raw accuracy gap on corrupted images, which conflates a pre-existing matched-clean gap with any corruption-specific interaction. We introduce a paired, floor-guarded protocol for executable TensorRT INT8 and FP8 treatments. Each clean or corrupted condition is materialized once and shared byte-for-byte across formats. The direct interaction is $\Delta E = (\text{FP8} - \text{INT8})_{\text{corrupt}} - (\text{FP8} - \text{INT8})_{\text{clean}}$, estimated with common image-bootstrap draws; absolute corrupted AP guards against apparent convergence at a shared failure floor. The exploratory grid contains 144 cells over four datasets, three YOLO11 capacities, four synthetic corruptions, and three ordinal severities. The raw corrupted gap favored FP8 in 134 cells, but the clean gap favored FP8 in all 144; clean adjustment changed the interpretation in 56 cells. FP8 retained 1.40 more matched-clean AP on average across 12 dataset–model blocks, whereas the balanced interaction was -0.02 AP points (95% percentile interval, -0.24 to +0.15). Untouched VOC/KITTI final partitions yielded -0.55 AP points ($[-0.78, -0.29]$). Absolute AP decreased in 282 of 288 corrupted format arms, with 35 below 10 AP. Architecture stress cases further show that a favorable interaction sign can follow a failed clean-fidelity gate. Clean fidelity, absolute corrupted accuracy, and corruption-specific interaction are therefore distinct deployment claims and should be reported together.

1. Introduction

Object detectors are increasingly deployed under simultaneous accuracy, latency, memory, and energy constraints. Post-training quantization (PTQ) reduces numerical precision without repeating full model training and is therefore attractive for rapid deployment[1, 8]. Integer arithmetic remains a standard path to compact inference, while FP8 provides an 8-bit floating-point alternative with a wider dynamic range and increasing accelerator support[21, 25]. These benefits matter most outside the laboratory, where images are also affected by sensor noise, motion, visibility loss, and lossy transmission. Common-corruption benchmarks and detector-specific studies show that such changes can substantially reduce recognition and localization accuracy[6, 18, 20].

Quantization and corruption are usually assessed with different summaries. Quantization papers emphasize a clean full-precision-to-quantized accuracy gap, whereas corruption papers emphasize an absolute clean-to-corrupted drop. A raw corrupted-image difference between two quantized engines answers neither question cleanly. If FP8 already exceeds INT8 on the matched clean input, the corrupted FP8–INT8 gap contains that pre-existing difference plus any additional format–corruption interaction. Calling the raw gap a robustness advantage therefore confounds clean fidelity with corruption-specific behavior.

A second ambiguity arises near an accuracy floor. Severe corruptions may drive both engines toward very low AP, causing their difference to contract. A signed interaction can then appear favorable precisely because neither engine remains useful. A defensible evaluation must therefore distinguish three claims: (i) matched-clean fidelity, (ii) absolute corrupted accuracy, and (iii) the change in the format discrepancy after the matched-clean difference has been removed. These claims are related, but none substitutes for the others.

The labels “INT8” and “FP8” also require care. An executable TensorRT engine is a treatment comprising number format, quantizer placement, calibration, graph rewriting, mixed-precision coverage, runtime tactics, preprocessing, and decoding. Two engines with the same nominal format can behave differently when any of these components changes. Our target is consequently the comparison of the recorded TensorRT INT8 and FP8 treatments, not an intrinsic ordering of all possible INT8 and FP8 implementations.

We introduce a paired, floor-guarded protocol that makes these distinctions explicit. For every clean or

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corrupted condition, images are materialized once and then consumed byte-for-byte by both quantized engines. The direct estimand is a four-cell difference in differences,

$$\Delta E = (\mathrm{FP8} - \mathrm{INT8})_{\mathrm{corrupt}} - (\mathrm{FP8} - \mathrm{INT8})_{\mathrm{clean}},$$

computed within common image-bootstrap draws. An accompanying absolute-AP guard prevents gap contraction near a shared failure floor from being labelled robustness. The exploratory landscape contains four datasets, three YOLO11 capacity rungs, four synthetic corruption families, and three ordinal severities, for 144 direct comparison cells. We then probe the result with untouched VOC/KITTI final partitions, alternative weights, repeated corruption materializations, training and calibration seeds, a fixed-class-universe bootstrap, measured engine runtime, and portability stress cases on RT-DETR and RetinaNet.

The main contributions are:

- a byte-matched four-cell estimand that isolates the corruption-induced change in the INT8–FP8 discrepancy while preserving image-level covariance;
- a floor-guarded reporting rule that jointly presents clean fidelity, absolute corrupted AP, and the clean-adjusted interaction instead of reducing them to one robustness score;
- a decision-impact analysis showing when the clean adjustment changes the practical interpretation of the raw corrupted gap; and
- a layered validation study that makes the conditional scope of split, realization, seed, class-universe, runtime, and architecture evidence explicit.

The empirical conclusion is deliberately narrower than a format ranking. FP8 preserves more matched-clean AP in every YOLO dataset–model block, but the corruption interaction is heterogeneous and changes with the evaluation scope. The contribution is therefore an evaluation methodology and an auditable body of treatment-conditional evidence, not a universal claim that one 8-bit format is more robust.

2. Related work

2.1. Post-training quantization for detection

Classical PTQ methods optimize scales, rounding, or local reconstruction to reduce the gap between full-precision and quantized networks [7, 14, 22]. Object detection is especially sensitive because classification and localization heads, feature pyramids, and highly imbalanced foreground/background responses must remain jointly calibrated. Fully Quantized Networks established low-bit detection as a distinct problem[13]; PD-Quant introduced a prediction-difference objective[17]; and DetPTQ selects layer-specific reconstruction metrics using an object-detection output loss [23]. More recently, InlierQ suppresses task-irrelevant activation anomalies while retaining informative volumes with a small, label-free calibration set[11]. These methods target clean quantization fidelity. Our work addresses the complementary question of how to evaluate quantized detectors when the input distribution is deliberately corrupted.

2.2. INT8, FP8, and executable treatments

Integer-only inference provides an established hardware path for efficient neural networks[8]. FP8 formats trade mantissa precision for exponent range and can preserve activation outliers that are difficult for a fixed integer scale[21]. FP8 PTQ studies have reported competitive accuracy and efficiency across model classes[25], while flexible-format work shows that the preferred 8-bit representation can vary across layers and architectures[30]. This literature motivates a direct comparison but also cautions against reducing an engine to its nominal datatype. Quantizer coverage, scale granularity, calibration, graph transformations, and unsupported operations define the actual treatment. We therefore inspect Q/DQ graph structure and interpret all results as conditional on the recorded ModelOpt/TensorRT pipelines.

2.3. Corruption robustness and quantized models

Common-corruption benchmarks established standardized synthetic shifts for image classification[6]. Detector studies then showed that localization systems can fail differently from classifiers under weather, noise, blur, and camera-motivated distortions [18, 20]. Work in *Image and Vision Computing* has also addressed adverse-weather detection, test-time adaptation, compact edge localization, and small-object failure modes [19, 26, 29, 32]. Small objects are particularly vulnerable because fewer pixels support class and box estimates[3], but low absolute small-object AP does not by itself imply a precision-specific small-object interaction.

Research directly joining quantization and distribution shift is emerging. Karimov et al. compare detector quantization under input degradations [10]; TTAQ adapts PTQ under continuous domain change [27]; and Recti-Q targets out-of-distribution robustness through feature-space rectification[28]. Related adversarial work finds that apparently inconsistent robustness conclusions can arise from changes in the quantization pipeline itself[12]. These studies reinforce the need to state the complete treatment and the exact contrast being estimated.

2.4. Positioning of the present study

Our work does not propose a new quantizer, corruption generator, or detector. It contributes a measurement protocol for executable engines. Three design choices distinguish it from a raw corrupted-accuracy comparison. First, the clean and corrupted inputs are codec-matched and shared byte-for-byte across formats. Second, the clean FP8–INT8 gap is removed inside a common image resample, yielding a direct difference in differences rather than a subtraction of separately summarized results. Third, the signed interaction is interpreted only after matched-clean fidelity and absolute corrupted AP have been checked. The resulting protocol is designed to be reusable across architectures and runtimes, including cases in which a PTQ recipe fails to transfer.

3. Methods

3.1. Study design and treatment definition

Figure 1 summarizes the study. COCO uses the official Ultralytics YOLO11n/m/x checkpoints; VOC, KITTI, and TT100K use recorded `best.pt` checkpoints trained within the same YOLO11 family. Each checkpoint is exported to an executable TensorRT ladder. FP16 is used only as a gross internal-consistency gate. The scientific arms are an FP32-typed TensorRT reference, entropy-calibrated INT8, and FP8. Clean and corrupted images are materialized before the precision branch, so both quantized engines receive the same ordered image IDs and the same encoded bytes.

The terms INT8 and FP8 denote the complete recorded ModelOpt/TensorRT treatments: quantizer placement, Q/DQ coverage, calibration list, scale granularity, graph rewriting, builder configuration, runtime, preprocessing, and decoder. They do not imply that every graph operation executes in the named datatype, and the study does not estimate the best achievable implementation of either format.

3.2. Estimands and interpretation order

Let d, m, q, b, and c index dataset, model, quantized treatment, size endpoint, and corruption condition. The symbol 95 denotes the matched clean condition: no named corruption followed by deterministic JPEG quality 95 with chroma subsampling disabled. All AP calculations use the native $[0, 1]$ scale; values are multiplied by 100 and reported as AP points.

The format-specific quantization gap is

$$Q_{d,m,q,b,c} = \mathrm{AP}_{d,m,\mathrm{FP32},b,c} - \mathrm{AP}_{d,m,q,b,c}. \tag{1}$$

The excess corruption gap is

$$E_{d,m,q,b,c} = Q_{d,m,q,b,c} - Q_{d,m,q,b,95}. \tag{2}$$

Positive E means that corruption widens the FP32-to-quantized gap beyond its matched-clean value; negative E means that the gap contracts.

For the direct comparison, define the clean and corrupted FP8–INT8 gaps

$$G_{d,m,b,x} = \mathrm{AP}_{d,m,\mathrm{FP8},b,x} - \mathrm{AP}_{d,m,\mathrm{INT8},b,x}, \quad x \in \{95, c\}. \tag{3}$$

The primary interaction is

$$\begin{aligned}\Delta E_{d,m,b,c} &= E_{d,m,\mathrm{INT8},b,c} - E_{d,m,\mathrm{FP8},b,c} \\ &= G_{d,m,b,c} - G_{d,m,b,95}.\end{aligned} \tag{4}$$

The FP32-typed reference cancels algebraically. Positive ΔE means that the FP8–INT8 AP gap is larger under corruption than on matched clean; negative ΔE means that the gap contracts. Neither sign states whether the corrupted accuracy is operationally useful.

The absolute corruption loss for precision treatment p is

$$D_{d,m,p,b,c} = \mathrm{AP}_{d,m,p,b,95} - \mathrm{AP}_{d,m,p,b,c}. \tag{5}$$

We call the protocol *floor-guarded*, rather than floor-corrected, because D and corrupted AP are interpretation checks; they do not transform or censor ΔE.

For COCO, VOC, and KITTI, the size interaction is

$$\Psi_{d,m,q,c} = E_{d,m,q,S,c} - E_{d,m,q,L,c}, \tag{6}$$

with direct contrast $\Delta\Psi = \Psi_{\mathrm{INT8}} - \Psi_{\mathrm{FP8}}$. TT100K uses an analogous original-box-height contrast. Area and height endpoints are reported as separate measurement families.

Table 1 states the required interpretation order. An engine that fails the application's matched-clean accuracy requirement is not rehabilitated by a favorable interaction sign. A signed interaction is then read together with absolute corrupted AP; efficiency is considered only after these accuracy checks.

3.3. Finite-grid target and evidence layers

The exploratory landscape contains 4 datasets $\times 3$ YOLO11 capacities $\times 4$ corruptions $\times 3$ severities $= 144$ direct cells. Its balanced macro is the arithmetic mean over this finite prespecified grid,

$$\overline{\Delta E}_{\mathrm{grid}} = \frac{1}{144} \sum_{d,m,c} \Delta E_{d,m,c}. \tag{7}$$

It is not a population-weighted effect for an unspecified deployment stream. Image-count, object-count, leave-one-dataset, and leave-one-corruption summaries are reported as alternative descriptive targets.

The evidence layers answer different validity questions (Table 2). The four-dataset grid is exploratory because VOC2007 test and the recorded KITTI validation partition participated in the checkpoint-selection workflow. A separate VOC/KITTI rerun uses untouched final partitions and retrained checkpoints; it is the strongest split-independent evidence for the stated conditional contrast, but it covers only two datasets. Seed, corruption realization, fixed-universe, and architecture studies are sensitivity analyses, not replications of one universal parameter.

3.4. Datasets and size endpoints

Table 3 summarizes COCO val2017[16], PASCAL VOC2007 test[4], the recorded KITTI validation split [5], and TT100K test[33]. VOC and KITTI annotations

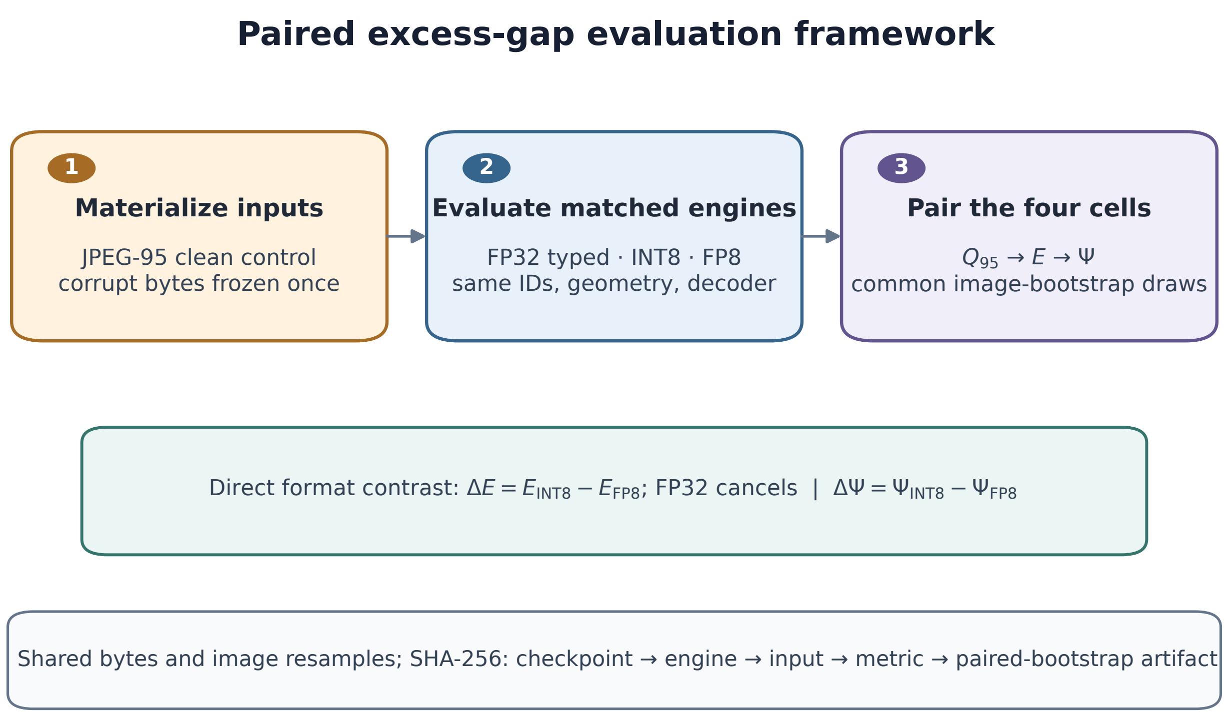

Figure 1: Paired evaluation protocol. A deterministic JPEG-95 matched-clean control and each corrupted input are materialized before the precision branch. The direct four-cell block crosses INT8/FP8 with clean/corrupted input and is resampled with common image draws. Matched-clean fidelity and absolute corrupted AP are checked before the clean-adjusted interaction is interpreted.

Table 1
Measurement contract. All linked contrasts share the checkpoint, ordered images, materialized bytes, decoder, and image-bootstrap draw.

Question	Quantity	Isolated comparison	Interpretation boundary
Matched-clean fidelity	Q_{95} or G_{95}	Accuracy cost before any named corruption	Does not establish corruption behavior.
Absolute corrupted accuracy	AP_c and D	Retained task accuracy and clean-to-corrupted loss	Guards against common convergence near an AP floor.
Format-specific interaction	$E = Q_c - Q_{95}$	Corruption-induced change in one quantized treatment's FP32 gap	Conditional on the recorded reference and treatment.
Direct format interaction	$\Delta E = G_c - G_{95}$	Change in the INT8–FP8 discrepancy; FP32 cancels	Not an intrinsic format ranking; report with clean and absolute AP.
Size interaction	$\Delta\Psi$	Difference between small-like and large-like format interactions	Area and height endpoint families remain separate.
Efficiency	engine size and latency	Runtime of the accepted executable treatment	Engine-only timing is not end-to-end deployment latency.

were immutably converted to COCO format. The reported values are therefore COCO-style measurements on derived annotations, not official VOC or KITTI challenge scores; VOC difficult-object and KITTI `DontCare`/difficulty semantics are not reproduced as leaderboard metrics.

TT100K contains 3,067 of the 3,071 referenced test images because four upstream files were unavailable. Its class map declares 221 categories, of which 136 occur in the retained annotations. COCO, VOC, and KITTI use original-pixel area strata: small $< 32^2$, medium $[32^2, 96^2)$, and large $\geq 96^2$ pixels. TT100K uses original bounding-box height: XS $[0, 12)$, S $[12, 24)$, M $[24, 48)$, L $[48, 96)$, and XL $[96, \infty)$ pixels. Its small-like endpoint combines XS + S and its large-like endpoint combines L + XL.

Table 2
Evidence hierarchy and permitted interpretation.

Layer	Scope	Uncertainty or dependency examined	Permitted claim
Exploratory landscape	144 YOLO11 cells; 2,000 paired image draws	Finite-image uncertainty and factorial heterogeneity	Conditional map and finite-grid macro
Untouched holdout	VOC and KITTI; six retrained checkpoints; 72 cells	Evaluation-after-selection reuse on two transfer datasets	Strongest split-independent conditional estimate
Training and calibration seeds	YOLO11m; three transfer datasets; severity 5	Recorded 3×3 seed crossing	Bounded descriptive seed sensitivity
Corruption realizations	36 base conditions; three frozen realizations	Stochastic materialization dependence	Bounded realization sensitivity
Fixed class universe	One TT100K stress cell; 10,000 weighted draws	Sparse category composition	One-cell endpoint sensitivity
Architecture portability	RT-DETR-L and RetinaNet; point estimates	Transfer of one frozen PTQ recipe	Failure-revealing stress evidence, not format superiority

Table 3
Frozen evaluation datasets. Counts refer to the converted annotations consumed by the evaluator.

Dataset	Split	Images	Objects	Declared/observed classes	Input px	Size strata
COCO	val2017	5000	36781	80/80	640	area: $< 32^2$ / $[32^2, 96^2)$ / $\geq 96^2$
VOC	VOC2007 test	4952	12032	20/20	640	area: $< 32^2$ / $[32^2, 96^2)$ / $\geq 96^2$
KITTI	validation	1496	8128	8/8	640	area: $< 32^2$ / $[32^2, 96^2)$ / $\geq 96^2$
TT100K	test	3067	8181	221/136	1280	height: XS/S/M/L/XL

3.5. Training, checkpoint selection, and holdout rerun

YOLO11n, YOLO11m, and YOLO11x provide three capacities within one architecture family[9]. VOC and KITTI use 640×640 input; TT100K uses 1280×1280 to retain more sign detail. The exploratory transfer checkpoints were trained for 100 epochs with seed 20260807, deterministic mode, automatic optimizer and batch selection, automatic mixed precision, eight data workers, and zero patience. The recorded augmentation used HSV amplitudes 0.015/0.7/0.4, translation 0.1, scale 0.5, horizontal-flip probability 0.5, and mosaic probability 1.0, disabled for the final ten epochs. MixUp, CutMix, copy-paste, rotation, shear, and perspective were zero. The trainer-selected `best.pt` defines every precision rung for a dataset–capacity block. COCO uses the corresponding official pretrained checkpoints without retraining.

The untouched rerun uses a prospectively frozen split registry (seed 20260818). For VOC, train2007 + train2012 provide 8,218 training images, val2007 provides 2,510 checkpoint-selection images, and val2012 provides 5,823 untouched final images. For KITTI, 1,496 images remain the selection partition; a deterministic, class-covered 1,197-image subset of the remaining records is reserved for final evaluation and 4,788 images are used for training. Every observed class occurs in each final partition. Calibration images are drawn only from the corresponding training split, and the confidence threshold, NMS, decoder, and calibration protocol are frozen before final evaluation.

3.6. Executable precision treatments and calibration

Engines were built with TensorRT 11.1.0.106 on an NVIDIA GeForce RTX 5090 using a 4096-MiB builder workspace. The FP32-typed reference does not explicitly disable TF32; "FP32" therefore names the implemented TensorRT reference and not strict IEEE FP32 execution for every kernel. This limitation does not affect the direct ΔE, because the FP32 term cancels in Eq. (4). FP16 is accepted only when its matched-clean AP differs from the FP32-typed engine by at most 1.0 AP point.

Calibration manifests contain deterministic 512-image selections from the training split and exclude evaluation images. The rebuilt COCO ladder uses a common list for INT8 and FP8. Quantized ONNX graphs were produced with the recorded ModelOpt `int8` and `fp8` modes and entropy calibration. Inspection of all 24 YOLO quantized graphs verified INT8 zero-point tensors for INT8 Q/DQ nodes and `FLOAT8E4M3FN` zero-point tensors for FP8 Q/DQ nodes. Activation scales are per tensor, weight scales use axis 0 (per output channel), and stored zero points are zero. Selected weights and activations are quantized; Sigmoid and other operations outside Q/DQ regions remain at runtime- selected precision. Supplementary Table S4 reports graph-level coverage.

The recorded software environment used PyTorch 2.11.0 + cu128, Ultralytics 8.4.95, ModelOpt 0.45.0 for the rebuilt COCO ladder, ONNX 1.21.0, NumPy 2.4.4, Pillow 12.2.0, OpenCV 5.0.0, NVIDIA driver 610.43.02, and batch size one. Older transfer registries omit some build-metadata fields, but their retained ONNX graphs expose the verified datatype, scale-axis, and zero-point semantics.

3.7. Synthetic corruptions and execution controls

The suite contains four selected synthetic image corruptions. Gaussian noise uses $\sigma/255 \in \{8, 20, 40\}$; linear motion-blur kernels use $\{5, 13, 23\}$ pixels; fog strength/radius pairs are $(0.18, 6)$, $(0.38, 12)$, and $(0.62, 20)$; and JPEG qualities are $\{70, 40, 15\}$. Severity labels 1, 3, and 5 are ordinal within a corruption family and are not physically comparable across families. In the exploratory grid, stochastic noise fields and blur angles are independently realized across severity labels because severity is part of the deterministic seed. Severity margins are therefore descriptions of the recorded materializations, not within-image causal dose–response effects.

Every output is deterministically encoded as JPEG quality 95 with chroma subsampling disabled, and dimensions are preserved. Each manifest records the ordered image IDs, per-file hashes, dimensions, and generator parameters. The matched-clean control applies the same materialization without a named corruption. Corruptions are never generated inside an inference worker. Letterbox geometry, inverse mapping, a confidence threshold of 10^{-3}, class-wise NMS at IoU 0.7, and a cap of 300 detections are held fixed. Run records bind the engine, decoder, annotations, class map, input manifest, image order, and prediction identity.

3.8. AP evaluation and paired uncertainty

AP is evaluated across IoU thresholds 0.50:0.05:0.95. The image is the resampling unit. For each dataset, a deterministic schedule of 2,000 image-index draws is reused across every linked clean/corrupted and INT8/FP8 arm. Repeated bootstrap positions are represented as repeated evaluator entries with unique replicate-local identities, preserving image multiplicity and stable score ordering. This common schedule retains the covariance that would be lost by subtracting independently bootstrapped format summaries.

Two-sided percentile intervals quantify finite-evaluation-image uncertainty conditional on the frozen checkpoints, engines, calibration lists, decoder, corruption materialization, and evaluation split. They do not propagate training-seed, calibration-sample, engine-build, dataset-selection, or corruption-population uncertainty. Cell-wise interval signs are exploratory and are not multiplicity-adjusted discoveries. The joint grid, area, and TT100K height endpoints are calculated within common draws using frozen balanced weights. Area and height endpoints are never pooled.

TT100K resamples can change the effective set of positive classes after zero-positive categories are omitted from an AP average. A targeted sensitivity for TT100K/YOLO11m/motion blur/severity 5 therefore uses 10,000 positive-weight Bayesian image-bootstrap draws and a category–endpoint universe fixed to the full 3,067-image annotation. This is a separate sensitivity estimand, not a replacement for the ordinary image bootstrap.

3.9. Sensitivity designs

Corruption realization. YOLO11m is evaluated on class-covering 512-image subsets of VOC, KITTI, and TT100K with realization seeds 202608181, 202608182, and 202608183. Within an image and realization seed, the stochastic field or angle is held fixed while severity changes its magnitude; JPEG is deterministic. The design contains 36 base conditions and 108 realization cells. A nested summary resamples the three realization labels and uses the paired image draw at the same replicate index. With three fixed realizations, this is a bounded sensitivity analysis.

Training and calibration seeds. For YOLO11m on VOC, KITTI, and TT100K at severity 5, training seeds 20260807, 20260813, and 20260814 are crossed with three deterministic 512-image calibration lists generated from the same seeds. Only the training and calibration-list seeds vary. The 3×3 crossing produces 54 engines and 108 direct dataset–corruption cells. Marginal means and ranges are descriptive; no population seed variance is inferred.

Weighting and omission. The same 144-cell ledger is summarized by equal-cell, image-count, and object-count weights, and after omitting each dataset or corruption family in turn. These analyses examine aggregation dependence without creating new inference runs.

3.10. Architecture portability stress cases

RT-DETR-L, an end-to-end transformer detector[2, 31], and RetinaNet-R50-FPN-v2, an anchor-based dense detector [15, 24], are evaluated on VOC, KITTI, and TT100K. Each family uses one deterministic 100-epoch transfer checkpoint per dataset and the same frozen 512-image train-only calibration identity within a dataset. The INT8 and FP8 construction recipes are transferred without family-specific retuning. The question is whether the recorded recipe ports to a different graph, not what an optimized family-specific recipe could achieve.

The extension contains 234 original-source-clean/corrupted metric cells and 18 separately materialized JPEG-95 matched-clean cells. The main sign convention is reconstructed from the compact ledgers:

$$E = (\mathrm{AP}_{\mathrm{FP32},c} - \mathrm{AP}_{q,c}) - (\mathrm{AP}_{\mathrm{FP32},95} - \mathrm{AP}_{q,95}). \quad (8)$$

The direct contrast remains $\Delta E = E_{\mathrm{INT8}} - E_{\mathrm{FP8}}$. Only point estimates are available, so ranges and means are descriptive. Engine latency uses three 30-s `trtexec` repetitions after a 5-s warm-up and excludes preprocessing, host transfer, decoding, and NMS.

3.11. Artifact integrity and analysis boundary

The external analysis and artifact-generation pipeline validates expected factorial grids, provenance links, and SHA-256 bindings before producing the CSV, JSON, LaTeX, and figure assets consumed by this manuscript. LaTeX does not independently verify hashes; it consumes those prevalidated outputs. The direct audit binds the paired components, common draw caches, absolute-accuracy ledger, and deployment records. A separate canonicalization script reconstructs all 144 cross-family interactions from the distributed corrupted and matched-clean ledgers and asserts the sign convention before generating the manuscript table. A second deterministic script derives the decision-impact figure and checks its 144-cell sign and AP-floor inventories. The source package includes these manuscript-level scripts and their audits.

AI-assisted coding and language tools were used to refactor manuscript-level validation scripts, check internal consistency, and revise the text. They did not select experimental conditions, generate detector predictions, or replace human interpretation. Every script operates on frozen ledgers, contains explicit cardinality and sign assertions, and was reviewed and rerun by the authors; the authors remain responsible for the analyses and claims.

4. Results

4.1. Execution integrity and matched-clean fidelity

All 12 prespecified FP16 consistency gates passed the 1.0-AP-point tolerance. The largest absolute FP32-typed–FP16 difference was 0.540 AP points (TT100K/YOLO11x), and the largest among the COCO/VOC/KITTI area-evaluated blocks was 0.253. These checks exclude gross disagreement within the recorded TensorRT ladder; they do not estimate engine-build variance or prove exact source-checkpoint parity.

On the JPEG-95 matched-clean inputs, FP8 achieved higher AP than INT8 in all 12 YOLO dataset–model blocks. The mean advantage was 1.40 AP points, ranging from 0.08 on TT100K/YOLO11x to 3.97 on KITTI/YOLO11m. This is the most consistent format difference in the study: the recorded FP8 treatment preserves more clean accuracy. It does not determine how the difference changes under corruption.

4.2. Clean adjustment changes the practical interpretation

The raw corrupted FP8–INT8 gap was positive in 134 of 144 cells and averaged 1.38 AP points. Read alone, this result would suggest an almost uniform FP8 advantage under corruption. The matched-clean FP8–INT8 gap, however, was positive in all 144 cells and averaged 1.40 points. After removing it, the mean ΔE was -0.02 points; 78 cell estimates were positive and 66 were negative. Most importantly, 56 cells had a positive raw corrupted gap but a negative clean-adjusted interaction (Fig. 2A). No cell changed in the opposite direction. The adjustment therefore changes the scientific claim in 39% of the grid, rather than merely rescaling the same ranking.

Figure 2B links the signed interaction to the lower corrupted AP of the two treatments. Eighteen cells place at least one format below 10 AP and five place at least one below 5 AP. These cells illustrate why clean adjustment and an absolute-AP guard answer complementary questions.

4.3. Exploratory 144-cell interaction landscape

Using common image-bootstrap draws, the balanced four-dataset $\overline{\Delta E}_{\mathrm{grid}}$ was -0.02 AP points with a 95% percentile interval of $[-0.24, +0.15]$. The point estimates were highly heterogeneous, ranging from -4.28 (KITTI/YOLO11m/motion blur/severity 5) to $+3.10$ AP points (KITTI/YOLO11n/fog/severity 5). Dataset means also opposed one another: TT100K was $+0.61$, whereas KITTI was -0.46 AP points (Table 4). Thus the near-zero macro is not evidence of homogeneous zero interaction.

The separate area $\Delta\Psi$ over COCO, VOC, and KITTI was -0.50 AP points with interval $[-1.18, +0.01]$. TT100K's height-based $\Delta\Psi$ was -1.38 points with interval $[-2.51, -0.14]$. Negative values mean that the INT8–FP8 interaction is more negative for the small-like endpoint than for the large-like endpoint under the stated sign convention; they do not establish that either format is robust on small objects. The full cell cloud and joint endpoints are shown in Fig. 3; the factorial sign reversals are visible in Fig. 4. Condition-level interval counts and factor margins are reported in Supplementary Tables S5–S6 and are not treated as multiplicity-adjusted discoveries.

4.4. Untouched VOC/KITTI holdout evidence

The untouched rerun retains 72 equal-weight dataset–model–corruption–severity cells. Its balanced direct interaction was -0.55 AP points ($[-0.78, -0.54, -0.29]$ for the 2.5th, 50th, and 97.5th percentiles). VOC was -0.43 $[-0.54, -0.29]$ and KITTI was -0.67 $[-1.13, -0.20]$ (Table 5). FP8 retained 1.60 more matched-clean AP on average and had a positive clean advantage in all six dataset–model blocks. The

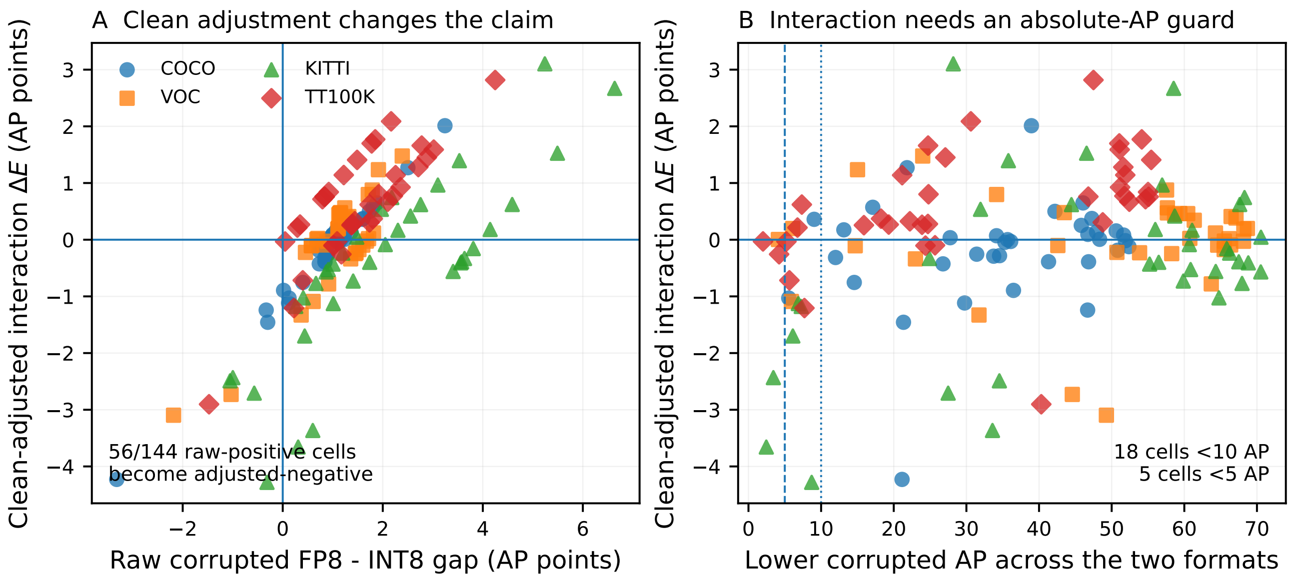

Figure 2: Decision impact of the protocol. (A) The raw corrupted FP8–INT8 gap includes the matched-clean gap; 56 of 144 raw-positive cells become negative interactions after clean adjustment. (B) The same interaction values are shown against the lower corrupted AP across INT8 and FP8. Dashed and dotted vertical lines mark 5 and 10 AP. Points are condition cells; colors and markers denote datasets.

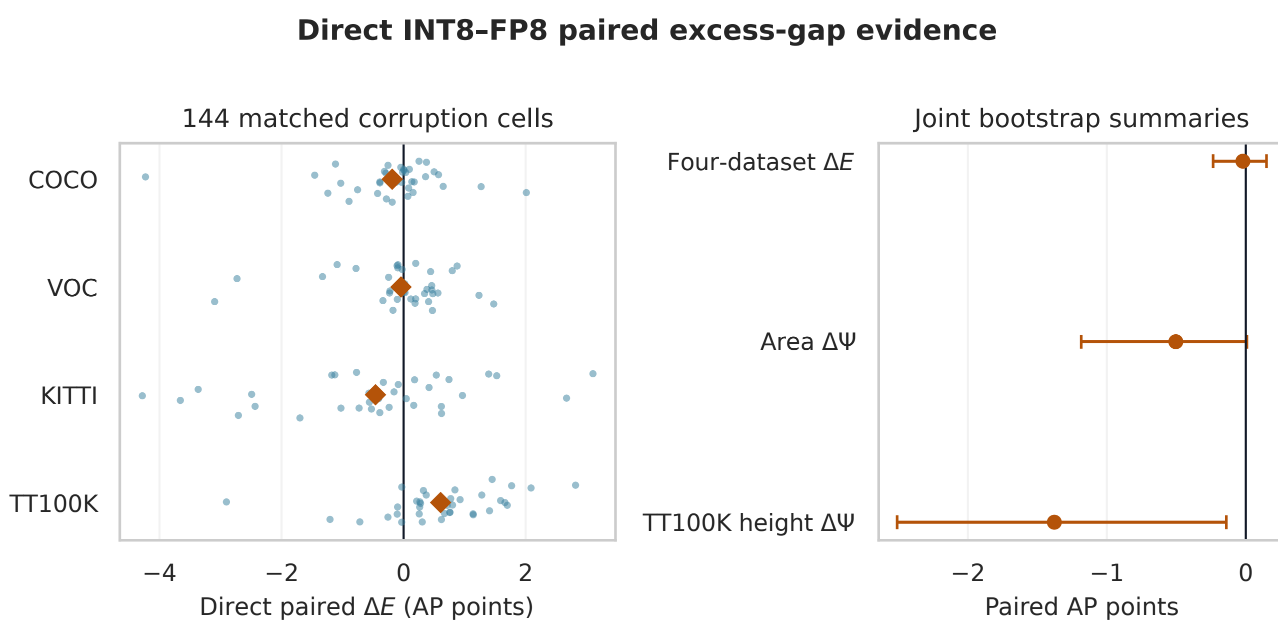

Figure 3: Exploratory direct interaction evidence. The left panel shows all 144 cell estimates and dataset means. The right panel shows joint paired-bootstrap summaries for the finite four-dataset grid and the two non-pooled size endpoint families.

size interaction was -0.13 AP points with interval $[-0.98, -0.33, +0.38]$.

This rerun removes evaluation-after-selection reuse for these two datasets and is therefore the strongest split-independent evidence in the paper. Its scope is nevertheless narrower than the exploratory four-dataset landscape. The two estimates answer different conditional questions and should not be pooled as replications of one population parameter.

4.5. Absolute corrupted accuracy and floor guard

Across the 288 format-specific corrupted YOLO arms, 282 had lower AP than their matched JPEG-95 clean arm. The median absolute loss was 5.51 AP points and the mean was 15.24, revealing a strongly right-skewed degradation distribution. Thirty-five arms fell below 10 AP and nine below 5 AP. In the untouched rerun, 18 of 144 corrupted format arms were below 10 AP and nine below 5 AP. These counts show that a

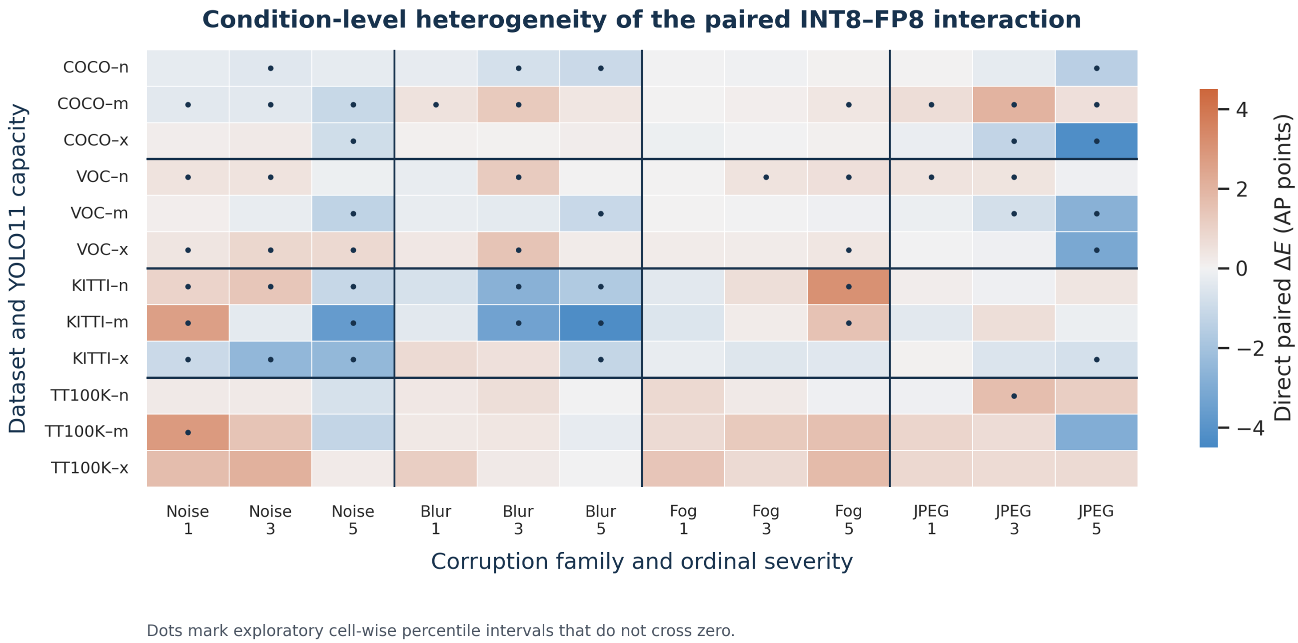

Figure 4: Condition-level heterogeneity. Columns group corruption family and ordinal severity; rows group dataset and YOLO11 capacity. Dots mark exploratory cell-wise percentile intervals that do not cross zero; they are not multiplicity-adjusted discoveries.

Table 4
Dataset margins from the exploratory direct ledger. Values are AP points; TT100K uses the height endpoint and the other datasets use area.

Dataset	Size endpoint	Cells	Mean ΔE	Mean $\Delta\Psi$
COCO	area	36	-0.19	-0.12
VOC	area	36	-0.04	-0.44
KITTI	area	36	-0.46	-0.95
TT100K	height	36	+0.61	-1.38

Table 5
Untouched-holdout dataset margins. Effects and intervals are AP points from the common paired schedule.

Dataset	Selection	Final	Cells	Mean ΔE	95% interval
VOC	2,510	5,823	36	-0.43	$[-0.54, -0.29]$
KITTI	1,496	1,197	36	-0.67	$[-1.13, -0.20]$

contracting format discrepancy can coexist with severe absolute failure.

Table 6 reports balanced dataset–format summaries. The all-corruption means hide substantial severity and condition variation, but they make the interpretation boundary explicit: ΔE describes a change in relative separation, not retained task performance. Size-specific absolute AP and floor counts are provided in Supplementary Table S2.

4.6. Sensitivity to weighting, realizations, seeds, and class universe

Alternative aggregation did not turn the small grid macro into a stable format ranking. Equal-cell, image-count, and object-count means were -0.02, +0.00, and -0.09 AP points, respectively, while the empirical cell SD was 1.20 points. Leave-one-dataset-out means ranged from -0.23 to +0.13; leave-one-corruption-out means ranged from -0.16 to +0.05. Sign changes around zero reflect conditional heterogeneity rather than contradictory arithmetic.

The three-realization analysis over 36 base conditions produced a mean ΔE of -0.23 AP points with nested percentiles $[-0.95, -0.28, +0.37]$. Mean between-realization variance was 0.3989 AP-points^2, compared with mean within-image-bootstrap variance of 0.7446. For $\Delta\Psi$, the mean was -1.04 with nested percentiles $[-3.05, -1.42, +0.10]$. With only three materializations, these values bound a recorded sensitivity rather than estimate a corruption population.

Across the targeted 108 training–calibration seed cells, the equally weighted mean ΔE was -1.11 AP points. Training-seed marginal means ranged from -1.64 to -0.68, and calibration-seed marginal means from -1.32 to -0.84. The original-seed intersection was -1.21, close to the balanced seed-grid mean, but the analysis covers only YOLO11m on three transfer datasets at severity 5.

For the fixed-universe TT100K stress cell, 10,000 positive-weight draws gave ΔE percentiles $[-1.19, -0.17, +0.88]$, compared with $[-1.79, -0.20, +1.34]$ under the ordinary 2,000-draw image bootstrap. The median sign and zero-crossing status were unchanged. The height-based $\Delta\Psi$ remained negative under both resampling schemes. Full sensitivity tables are reported in Supplementary Sections S7–S10.

Table 6
Absolute-accuracy guardrail from the validated YOLO metric ledger. $D = \mathrm{AP}_{95} - \mathrm{AP}_c$; rows are balanced descriptive summaries, not pooled leaderboard scores.

Dataset	Format	Clean AP	All-corrupt AP	Mean D	Severity-5 AP	Severity-5 D
COCO	INT8	46.010	34.911	11.099	25.185	20.826
COCO	FP8	47.110	35.824	11.286	25.667	21.443
VOC	INT8	65.732	49.246	16.486	35.478	30.255
VOC	FP8	66.828	50.300	16.529	36.023	30.805
KITTI	INT8	67.284	46.445	20.839	29.881	37.403
KITTI	FP8	69.797	48.498	21.299	31.504	38.293
TT100K	INT8	44.760	32.270	12.490	22.573	22.188
TT100K	FP8	45.636	33.752	11.884	23.467	22.169

Table 7
RTX 5090 TensorRT engine measurements. Condition values are medians of three raw-log-validated invocations; rows are medians across 12 matched conditions.

Precision	Conditions	Median engine (MiB)	Median latency (ms)	Median throughput (qps)
FP32	12	79.90	2.434	410.6
INT8	12	24.07	0.789	1265.3
FP8	12	21.92	0.760	1315.7

4.7. Engine-only runtime

Across the 12 matched YOLO dataset–model conditions, the median FP32-to-INT8 latency ratio was 2.96× (range 1.49–3.14×), and the FP32-to-FP8 ratio was 3.13× (1.78–3.36×). The matched INT8-to-FP8 ratio had a median of 1.08× and ranged from 0.92 to 1.19×; FP8 was therefore not faster in every condition. Table 7 reports engine size, latency, and throughput. These measurements exclude image decoding, preprocessing, transfer, detector decoding/NMS, power, and energy.

4.8. Architecture portability stress cases

FP8 remained within 1.15 matched-clean AP points of the FP32-typed engine in all six RT-DETR/RetinaNet dataset–family blocks. The transferred INT8 recipe did not preserve clean accuracy: its mean Q_{95} was 38.66 AP points for RT-DETR-L and 22.53 for RetinaNet. The largest failures were VOC/RT-DETR-L (69.30) and KITTI/RetinaNet (43.30). Two FP32-typed transfer blocks were themselves weak: RT-DETR-L/KITTI achieved 12.29 matched-clean AP and RetinaNet/TT100K 3.37. These blocks are retained to expose transfer failure rather than excluded after inspection.

Under the canonical $E = Q_c - Q_{95}$ sign, the mean INT8 interactions were -6.81 for RT-DETR-L and -6.21 AP points for RetinaNet. Mean corrupted INT8 AP was only 1.97 and 11.34, with 34/36 and 16/36 cells below 5 AP, respectively. FP8 mean E was near zero (-0.02 and $+0.01$) after much smaller clean gaps, with mean corrupted AP of 33.30 and 26.91. The descriptive direct contrasts were -6.79 and -6.23 AP points. Figure 5 shows why these negative values are not evidence of INT8 robustness: they largely describe contraction of a failed clean discrepancy near low absolute AP.

Both quantized treatments reduced engine-only latency relative to their family FP32-typed reference. Median RT-DETR speedups were 2.47× for INT8 and 2.30× for FP8; RetinaNet speedups were 4.56× and 3.94×. The faster INT8 medians do not compensate for the clean-fidelity failure. The supported conclusion is that the frozen recipe did not transfer, not that INT8 is intrinsically incompatible with either detector family.

5. Discussion

5.1. The protocol changes the question, not only the scale

The clearest methodological result is the 56-cell interpretation change. A raw corrupted FP8–INT8 gap favors FP8 in 134 of 144 cells, but the clean treatment already favors FP8 in every cell. After that baseline discrepancy is removed, 66 interactions are negative and the balanced mean is near zero. The adjustment therefore prevents a clean-fidelity result from being relabelled as a corruption-robustness result. This distinction is consequential even when the same engine, images, and AP evaluator are used: the choice of contrast changes what can legitimately be concluded.

The direct formulation also improves statistical efficiency and clarity. The four treatment arms remain inside each image draw, retaining the covariance induced by common images, bytes, preprocessing, and decoder. Subtracting two independently summarized format effects would discard that pairing and obscure which uncertainty sources are shared. The algebraic cancellation of the FP32-typed reference further isolates the treatment comparison from the TF32 status of that historical reference, while the format-specific Q and E decompositions remain available for diagnosis.

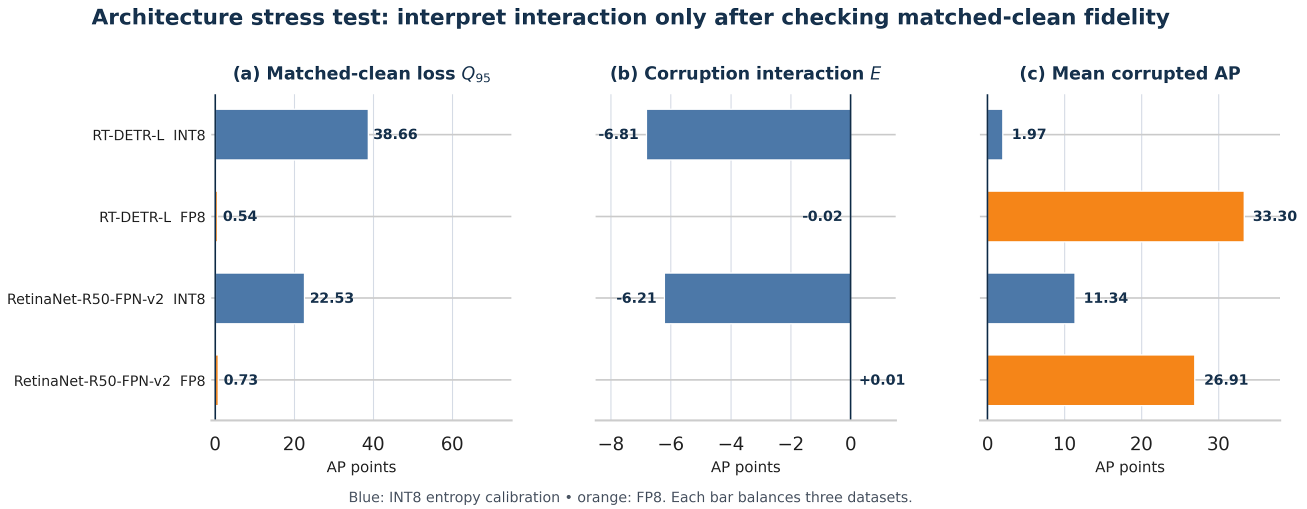

Figure 5: Architecture portability stress cases. Matched-clean loss is checked first, interaction second, and absolute corrupted AP third. Each bar balances VOC, KITTI, and TT100K. The negative INT8 interaction follows a large clean accuracy failure and low corrupted AP.

5.2. Clean fidelity, absolute accuracy, and interaction are distinct

FP8's 1.40-AP matched-clean advantage is consistent across all 12 YOLO blocks. That result is stronger and more uniform than the corruption interaction. The exploratory grid mean is -0.02 AP with an interval crossing zero, whereas the untouched VOC/KITTI estimate is -0.55 with an interval below zero. A negative ΔE means that the FP8–INT8 gap contracts relative to matched clean; it does not mean that INT8 retains more useful corrupted accuracy or that FP8 is fragile in an absolute sense.

The absolute guardrail is therefore not optional. AP falls in 282 of 288 corrupted YOLO arms, and 35 arms are below 10 AP. The architecture stress cases make the same point more sharply: the transferred RT-DETR INT8 recipe begins with a 38.66-AP mean clean gap, then shows a negative interaction while retaining only 1.97 mean corrupted AP. The interaction sign is arithmetically correct but operationally unhelpful without the preceding fidelity and floor checks. For deployment, the evidence should be read in order: matched-clean accuracy, absolute corrupted accuracy, interaction, and only then latency or engine size.

The term *floor-guarded* is intentionally modest. We do not propose a formal correction for bounded AP or a thresholded primary estimand. Instead, we make failure floors visible and prevent the signed contrast from carrying a claim it cannot support. Applications can add predeclared minimum-AP or maximum-loss gates appropriate to their risk tolerance; such gates should be specified before the interaction is inspected.

5.3. Why the exploratory and holdout estimates differ

The four-dataset landscape and the untouched VOC/KITTI rerun have different conditional scopes. The broader grid includes COCO and TT100K, three capacities, and evaluation partitions that were not uniformly independent of checkpoint selection. It maps heterogeneity across a finite factorial design. The holdout rerun removes evaluation-after-selection reuse for VOC and KITTI and retrains six checkpoints under frozen split registries. Its -0.55-AP estimate is the strongest split-independent evidence available here, but it does not cover COCO, TT100K, additional training seeds, or alternative detector families.

The difference between -0.02 and -0.55 should therefore not be treated as a failed replication. It shows that the interaction depends on the target regime. This interpretation is consistent with the exploratory dataset margins, leave-one-factor summaries, and 1.20-AP cell dispersion. Reporting one pooled number without its finite-grid definition would hide that dependence. A future confirmatory study should predeclare the deployment-weighted target, construct untouched partitions for every domain, and propagate training, calibration, engine-build, and acquisition uncertainty in one hierarchical design.

5.4. Heterogeneity and object size

The near-zero grid mean is produced by opposing condition effects rather than a uniform absence of interaction. Dataset margins range from -0.46 on KITTI to +0.61 on TT100K, and cell estimates span more than seven AP points. Alternative weights and leave-one-factor summaries remain small but cross zero. This pattern supports treatment- and regime-specific reporting rather than a single format robustness rank.

The size endpoints also resist a simple narrative. The area-based $\Delta\Psi$ interval reaches zero, whereas the TT100K height endpoint is negative. These quantities compare the format interaction between small-like and large-like objects; they do not measure absolute small-object resilience. Size-specific AP is often low, particularly for TT100K XS instances, so floor contraction can be stronger for the small endpoint. In addition, area thresholds and original-box-height thresholds define different scientific objects and must not be pooled. The results therefore do not support a universal claim that quantization-corruption interaction is amplified on small objects.

5.5. Treatment-level interpretation of INT8 and FP8

The graph audit shows that both nominal formats are mixed-precision treatments with architecture-dependent Q/DQ coverage. Calibration, scale granularity, operator exclusions, TensorRT tactic selection, output parameterization, and decoder behavior all contribute to the executable system. This treatment-level view explains why the same nominal INT8 recipe can preserve YOLO accuracy yet collapse on RT-DETR or RetinaNet. The portability stress test is valuable precisely because it exposes failure rather than because it supplies a fair cross-family leaderboard.

The evidence does not localize the cross-family INT8 collapse to a specific layer, quantizer, runtime stage, or decoder. Repair would require output-level parity checks, layer-wise sensitivity, alternative calibration, mixed-precision rescue, quantizer-search methods, or quantization-aware training. Conversely, FP8's close clean tracking across the six stress blocks is encouraging but remains conditional on one recipe, one hardware/software stack, one training seed, and one calibration list per dataset. Neither result establishes an intrinsic property of the number format.

5.6. Practical reporting recommendations

For an executable detector comparison under corruption, we recommend the following minimum report. First, define a matched-clean materialization and show the clean format gap. Second, materialize each corruption independently of precision and share the exact bytes across treatments. Third, estimate the clean-adjusted format interaction inside common image resamples. Fourth, report absolute corrupted AP or clean-to-corrupted loss, with any application-specific accuracy gate declared in advance. Fifth, retain condition-level or stratified results so that a balanced mean cannot conceal reversals. Finally, state the full executable treatment and separate engine-only timing from end-to-end latency.

These recommendations extend beyond INT8 and FP8. The same four-cell structure can compare quantizers, mixed-precision policies, compilers, calibration strategies, or hardware runtimes whenever a pre-existing clean discrepancy may be confused with a distribution-shift interaction.

5.7. Limitations

Several limitations bound the conclusions. The corruption suite contains four selected synthetic transformations and does not represent the distribution of natural weather, sensor, compression, or domain shifts. Severity is ordinal and, in the exploratory grid, stochastic realizations are not nested across levels. The three-realization analysis is too small to estimate a population of possible fields or angles.

The exploratory macro is a finite-grid arithmetic mean, not a deployment- weighted population effect. The broad grid remains conditional on one primary training seed and recorded checkpoint-selection workflows; the untouched rerun covers only VOC and KITTI. The targeted seed crossing fixes YOLO11m, three transfer datasets, and severity 5, and does not estimate full training or calibration variance. Engine builds were not repeated, and the bootstrap intervals condition on the frozen executable artifacts.

VOC and KITTI are evaluated through converted COCO-style annotations rather than official challenge protocols. TT100K's sparse class universe can vary under ordinary image resampling; the fixed-universe check covers only one stress cell. The area and height endpoint families are not interchangeable. The FP32-typed TensorRT reference may use TF32, although this does not affect the direct ΔE algebra.

Runtime is measured on one RTX 5090/TensorRT stack and excludes preprocessing, transfer, decoding, NMS, memory, power, and energy. Graph-level Q/DQ counts do not reveal TensorRT kernel precision or fallback execution. The cross-family extension uses point estimates, one training seed, one calibration identity, and fixed recipes that were not optimized per architecture. Finally, the compact submission package reproduces manuscript-level summaries from frozen ledgers but does not redistribute datasets, checkpoints, engines, or raw prediction payloads.

6. Conclusion

A corrupted-image difference between quantized detectors is not, by itself, a robustness comparison. It combines any pre-existing clean-fidelity gap with the additional change induced by corruption. We introduced a byte-matched, image-paired, floor-guarded protocol that separates these quantities through the direct interaction $\Delta E = (\mathrm{FP8} - \mathrm{INT8})_c - (\mathrm{FP8} - \mathrm{INT8})_{95}$ and interprets it only alongside matched-clean and absolute corrupted AP.

The distinction changes the evidence. Although the raw corrupted gap favored FP8 in 134 of 144 YOLO cells, clean adjustment changed the interpretation in 56 cells. FP8 preserved 1.40 more matched-clean AP on average across all 12 YOLO blocks, whereas the balanced exploratory interaction was -0.02 AP points with a 95% interval of $[-0.24, +0.15]$. Untouched VOC/KITTI final partitions yielded -0.55 [−0.78, −0.29], demonstrating a narrower but stronger split-independent conditional result. Absolute accuracy fell in 282 of 288 corrupted arms, with 35 below 10 AP, so gap contraction cannot be equated with retained performance. Architecture stress cases further showed that a negative interaction can arise when an INT8 recipe already fails its matched-clean fidelity gate.

The supported conclusion is not a universal INT8–FP8 robustness ranking. Clean fidelity, absolute corrupted accuracy, and the corruption-specific format interaction are different deployment claims. Reporting them together, under an explicit finite target and a fully specified executable treatment, produces a more auditable and decision-relevant evaluation of quantized computer-vision systems.

Data and code availability

The source package and Supplementary Data S2 supplied with this submission contain the LaTeX source, generated CSV/JSON ledgers, validation audits, and deterministic scripts needed to reproduce the manuscript-level summaries, cross-family sign checks, and decision-impact figure. Raw benchmark datasets, training checkpoints, TensorRT engines, and raw prediction payloads are not redistributed because of licensing and size constraints. The retained run and metric records bind the reported summaries to those artifacts by SHA-256. No public archival DOI is claimed in this version.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During manuscript preparation, the authors used OpenAI Codex, OpenAI ChatGPT, and Anthropic Claude Code for language revision, LaTeX drafting, code refactoring, and consistency checks. The tools did not autonomously select experimental conditions or generate detector measurements. The authors reviewed and reran the analysis and validation scripts, verified the numerical claims and citations, edited the resulting text, and take full responsibility for the content of the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Dinh Thuan Nguyen: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing–original draft. **Lam Phuong Nguyen:** Validation, Data curation, Writing–review and editing. **Vinh Huy Nguyen:** Methodology, Visualization, Writing–review and editing. **Mohan Rajesh Elara:** Supervision, Writing–review and editing. **Anh Vu Le:** Supervision, Project administration, Writing–review and editing.

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